

CS23502	NETWORKS AND DATA COMMUNICATION	L	T	P	C
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UNIT – I INTRODUCTION TO NETWORKING AND APPLICATION LAYER 7L, 16P

Building a network, Network edge and core – Layered Architecture, ISO/OSI Model, Internet Architecture (TCP/IP) – Networking Devices: Hubs, Bridges, Switches, Routers, and Gateways – Performance Metrics – Introduction to Sockets – Application Layer protocols – HTTP/HTTPS – FTP/ SFTP – Email – DNS – DNSSEC.

PRACTICALS:

1. Applications using TCP sockets:
 - a) Echo client and echo server
 - b) Chat

UNIT – II TRANSPORT LAYER 9L, 12P

Introduction – Connectionless Transport: User Datagram Protocol – Principles of Reliable Data Transfer (GBN, SR) – Connection– Oriented Transport – TCP – Connection establishment and teardown – Triggering transmission – Flow Control – Congestion Control – Transport Layer Security – TLS – SSL

PRACTICALS:

1. Write socket programs to simulate the operation of the following application layer protocols:
 - a) HTTP and Web caching
 - b) DNS

UNIT – III NETWORK LAYER 11L, 16P

Inside a Router – Internet Protocols – IPV4, IPV6, IP Addressing and NAT – Subnetting – Variable Length Subnet Mask (VLSM) – Classless Inter– Domain Routing (CIDR) – Routing Algorithms – Distance Vector Routing – Link State Routing – RIP – OSPF – BGP – ICMP – DHCP

PRACTICALS:

1. Analyze the Network traffic using Packet Analyser (Wireshark) and understand the various protocol headers.
2. Simulation of flow control
2. Practice different network commands available in Windows and Linux Operating Systems and troubleshoot the network.

UNIT – IV LINK LAYER AND PHYSICAL LAYER 9L, 8P

Introduction – Link Layer Framing, Addressing – Error Detection/ Correction Techniques – Switched Local Area Networks (ARP, Ethernet, VLAN) – Wireless LAN (802.11) – Physical Layer: Signals – Bandwidth and Data rate – Encoding – Multiplexing – Transmission media

PRACTICALS:

1. Configure the network devices such as Router, Switch, Hub, Bridge, and Repeater by simulation.
2. Simulation of Distance Vector/ Link State Routing algorithm

UNIT – V SDN AND NFV**9L, 8P**

SDN: Background and Motivation – Evolving Network Requirements – SDN Architecture – SDN Data Plane and OpenFlow – SDN Control Plane Architecture – Virtual Machines – NFV Concepts – NFV Benefits and Requirements

PRACTICALS:

1. Performance evaluation of routing protocols using the simulation tool
2. Configuring client– server interaction using mininet with an SDN controller

TOTAL: 45L + 60P = 105 PERIODS**COURSE OUTCOMES :**

Upon completion of the course, the students will be able to

1. Highlight the significance of the functions of each layer in the network
2. Identify the devices and protocols to design a network and implement it
3. Build network applications using the right set of protocols and estimate their performance
4. Explain media access and communication techniques
5. Illustrate the techniques behind SDN/NFV

REFERENCES:

1. James F. Kurose, Keith W. Ross, "Computer Networking: A Top– Down Approach", Eighth Edition, Pearson Education, 2022.
2. Larry L. Peterson, Bruce S. Davie, "Computer Networks: A Systems Approach", Sixth Edition, Morgan Kaufmann Publishers Inc., 2021.
3. William Stallings, "Foundations of modern networking: SDN, NFV, QoE, IoT, and Cloud", 1st edition, Addison– Wesley Professional, 2015.
4. William Stallings, "Data and Computer Communications", Tenth Edition, Pearson Education, 2017.
5. Ying– Dar Lin, Ren– Hung Hwang, Fred Baker, " Computer Networks: An Open Source Approach", 1st Edition, McGraw Hill, 2011

CO– PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	1	1	1	–	–	–	–	–	3	2	2
CO2	3	3	3	3	2	1	1	–	3	–	–	2	3	3	2
CO3	3	3	3	3	2	1	1	–	3	–	–	2	3	3	3
CO4	3	3	3	2	1	1	1	–	1	–	–	1	3	1	1
CO5	3	3	3	2	2	1	1	–	–	–	–	2	3	2	3

1– low, 2– medium, 3– high, '–' – no correlation